OPERATING INSTRUCTION MANUAL

MODEL TM-620

TEMPERATURE MONITOR

WARNING!

DO NOT ATTEMPT TO OPERATE THIS EQUIPMENT BEFORE THOROUGHLY READING THIS INSTRUCTION MANUAL.

Table of Contents

Tables

1. Introduction

The TM-620 Temperature Monitor is a state-of-the-art instrument which provides excellent accuracy and resolution. Its versatile architecture allows operation with a variety of RTDs.

The TM-620 is available in dual and quad channel versions, referred to as the TM-622 and TM-624, respectively. It supports fixed constant current excitations of 1mA, 100uA, and 10uA, with a voltage compliance level of 5V and a measurement range of 0-5V. Measurements are presented as temperature (degrees Kelvin, Celsius, or Fahrenheit), ohms, or volts. Temperature readings require temperature curves to be loaded onto the device. The TM-620 has room for six user-loadable temperature curves of up to 200 points each, in addition to pre-loaded curves for RO600 and PT100 RTDs.

The TM-620 has user-adjustable high and low setpoints which can be used to trigger an audible alarm and control other equipment via internal relays. If the measured temperature is between the high and low setpoints the alarm is silent. If temperature moves below the low setpoint or above the high setpoint, the alarm sounds until silenced by the operator. Each channel also provides 4-20mA outputs which track the current temperature reading.

Computer control via USB and Ethernet is available in the standard configuration, with GPIB available as an optional add-on.

Note that “TM-620” is used in place of “TM-622” or “TM-624” except for when distinction is necessary.

2. Factory Calibrations, Installed Options and Certification

Front Panel:

Model: **TM-622 / TM-624**

Serial Number: ____________________________

Internal Reference Number: ___________

Firmware Version (Panel/Ch): ___________

RTDs:

Number of Provided RTDs: ________

	RTD 1	**RTD 2**
Manufacturer/Serial #	______________	______________
Type:	______________	______________
Excitation:	______________	______________

Notes: ____

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Certified: ____________________________

Date: ____________________________

3. General Information

The TM-620 is delivered fully calibrated, tested, and ready to operate. This includes pre-loaded temperature curves if Cryomagnetics provides RTDs with the device.

3.1. Line Voltage and Input

The TM-620 operates using 110-240VAC at 50/60 Hz. Its power input contains an EMI/RF line filter, short circuit protection, overvoltage protection, and overcurrent protection.

3.2. RTDs

RTDs connect to the TM-620 through the rear panel DB-9 connector(s). Ensure that the device is powered off while connecting or disconnecting an RTD. See section 5.1 for pinouts.

3.3. Analog Outputs

Relay pins and 4-20mA outputs are presented on the rear panel male Mini XLR 8 pin connector. See section 5.2 for pinouts. Ensure that the device is powered off when connecting or disconnecting the analog connectors.

3.4. Calibration

TM-620 Calibration is performed at the factory. No calibration by the user is necessary.

3.5. Initial Setup

To use the TM-620, the user first needs to connect an RTD to the TM-620. Please make sure that power is off while connecting/disconnecting cables. I+ and V+ should connect to one leg of the RTD, I- and V- should connect to the other. **Ensure that I+/I- and V+/V- are not shorted, and that the I/V polarities are the same for each connection. Failure to do this can cause the device to malfunction.** Complete any connections to the analog output connector at this time as well.

After making all connections, turn on the device by pressing the power button located on the bottom right corner of the front panel. The device will immediately begin displaying readings. If an “Open Sensor” message appears, power down the device and double check all connections. Contact Cryomagnetics for assistance if this issue persists.

Once powered on, the device should be allowed to warm up for at least 15 minutes to let component and board temperatures settle out. Measurements are compensated to account for drifts in temperature over time.

4. Operation and Menus

The TM-620's normal operating display will show the measurements for each active channel, up to a total of four. Alarms and Local/Remote mode indicators will display on the top-level screen. If a subchannel is disabled, it will display as "Disabled." If the measured value of an RTD lies outside of its temperature curve, its reading will display as "----." If the 0V-5V input range of the device is surpassed, an "OVRRNG" warning will display.

Setup of the TM-620 can be performed either through the front panel keypad and display or through remote computer interface. To change settings, press the **Menu** key. Use the left and right arrow keys to select a tab and the down arrow key to enter that tab's fields. The keypad can be used to modify numerical fields in the main menu. All other fields use the Enter key to cycle through preset values.

Pressing **Menu** while in the main menu returns the instrument to normal display and operations. If any changes were made, the device will ask the user if he or she wants to keep the modified settings. Alternately, pressing the **Esc** key while in the main menu will return to the top-level display. Be aware that some computer interface commands are not available while in the menu (see Appendix A: Computer Interface Command Reference).

Each of the two available channels supports two subchannels for a maximum total of four different inputs. Subchannels A and B belong to channel 1, subchannels C and D belong to channel 2. Settings for subchannels A and B are located on one tab, while settings for subchannels C and D, if installed, are located on a separate tab.

4.1. Channel Tab

4.1.1. Units

The Units field lets the user select the display units of the TM-620. Available options are Kelvin, Celsius, Fahrenheit, Ohms, and Volts.

4.1.2. Excitation

Select a current excitation level for each subchannel. Options are 1mA, 100uA, 10uA, and "Disable."

4.1.3. Sensor Curve

The Sens. Curve setting allows the user to select from one of eight temperature curves: six custom curves plus pre-installed RO600 and PT100 curves. Custom curve names default to “Curve X,” where X corresponds with the curve number, from 1 through 6. Custom curve names can be added when loading curve data onto the device.

4.1.4. Alarm High/Low

The TM-620 has an internal audible alarm and per-channel alarm relays which can be set to trigger when the measured temperature (in Kelvin) falls outside of user defined setpoints. Alarm values are only checked when display units are set to temperature; alarm values will not be checked when units are set to volts or ohms. “A-LO” or “A-HI” will appear to the right of the corresponding channel’s reading on the top-level display if the alarm value is surpassed. The audible alarm can be silenced by pressing any front panel key on the TM-620.The visual indication of the alarm condition is maintained as long as the measured temperature remains outside of the alarm set points. Valid alarm values are 0K to 1000K. If alarm functionality is not required, the high and low levels can be set to values outside of the expected range of that channel.

4.2. Setup Tab

4.2.2. Interface

The TM-620 contains USB and LAN interfaces, plus an optional GPIB interface, all of which are selectable using the Interface tab. After changing interfaces, the device should be restarted. See sections 5.4, 5.5, and 5.6 for more information on interfaces.

4.2.3. Baud Rate

This field allows the user to select baud rates from 9600 to 115200 for use with the onboard RS-232 to USB adapter.

4.2.4. IP Address

This field allows the user to set the IP address of the device in LAN mode.

4.2.5. Subnet

This field allows the user to set the subnet address of the device in LAN mode.

4.2.6. Gateway

This field allows the user to set the gateway address of the device in LAN mode.

4.2.7. MAC Address

This field is not editable and shows the MAC address of the device.

4.2.8. Socket

This field allows the user to set the port of the device in LAN mode.

4.2.9. Alarm

Allows for the audible alarm to be enabled or disabled.

5. Interfaces

The TM-620 comes standard with both LAN and USB interfaces, plus analog outputs for additional monitoring/control. An optional GPIB interface is also available.

5.1. RTD Connector

The constant current excitation and voltage measurement pins are located on the rear panel DB9 connector(s) of the TM-620. The TM-620 is pin-compatible with the TM-612 and TM-614. It is highly recommended that sensors be connected using shielded, twisted pair wire in a four-wire configuration. Failure to do so may lead to decreased performance and increased susceptibility to noise. The shell of the rear panel DB9 connector is connected to earth ground. The RTD pinout is as follows:

Pin #	Description	Pin #	Description
1	V_B+ / V_D+	6	I_B+ / I_D+
2	V_B- / V_D-	7	I_B- / I_D-
3	NC	8	I_A+ / I_C+
4	V_A+ / V_C+	9	I_A- / I_C-
5	V_A- / V_C-		

Table 1: RTD 4 Wire Connections

Note that subchannels A/B correspond with channel 1 and subchannels C/D correspond with channel 2.

5.2. Analog Outputs

Analog outputs from the TM-620 are located on the male Mini XLR 8 pin connector on the rear panel. This connector is intended to be mated with a TA8FLX female plug or equivalent. The 4-20mA output signals are scaled so that the Alarm Low value corresponds with 4mA and the Alarm High value corresponds with 20mA. Values beyond these Low/High values will be limited to 4mA/20mA respectively. For example, given an Alarm Low value of 40K, an Alarm High value of 200K, and a reading of 100K, the output will be 10mA. The 4-20mA output has a compliance level of 12V. The alarm relays are normally open, and close when an alarm condition is detected. The relays are rated for a max load voltage of 60V and a max load current of 100mA.

Pin #	Description	Pin #	Description
1	Alarm B/D	5	Alarm A/C
2	4-20mA+ B/D	6	4-20mA+ A/C
3	Alarm B/D	7	4-20mA- A/C
4	4-20mA- B/D	8	Alarm A/C

Table 2: Analog Output Connections

Note that subchannels A/B correspond with channel 1 and subchannels C/D correspond with channel 2.

5.3. Audible Alarm

The TM-620 contains a user enabled audible alarm. The alarm may be set through the front panel menu system (see section 4.1.4). The alarm gives the user an audible indication that the measured temperature has fallen outside the alarm set points. Once the alarm is activated, it may be silenced by any front panel keystroke. It will only be reactivated after a non-alarming reading (i.e., a reading between the alarm setpoints) is detected by the TM-620, followed by another reading in which an alarm condition occurs again. The audible alarm may be disabled in the Setup menu.

5.4. LAN

The LAN interface requires the host PC to be on the same gateway as the TM-620. This can be done by setting a static IP address for the PC. Putty or Labview are recommended for interfacing over LAN.

5.5. USB

The USB port is accessed through the USB Type B connector on the rear panel of the instrument. To use the USB interface, a virtual COM port (VCP) driver must be installed on the host PC, which treats the USB interface as a traditional RS-232 interface. The virtual com port driver is available from Cryomagnetics or from FTDI. The TM-620 must be configured for the same baud rate as the PC's virtual COM port. The baud rate may be set to 9600, 19200, 38400, 57600 or 115200. Factory default is 9600. TeraTerm, Labview, or Python are recommended for serial interfacing.

5.6. GPIB

The optional TM-620 GPIB interface implements SH1, AH1, T6, L4, SR1, RL0, RL1, PP0, DC1, DT0, C0, and E1 options. The commands are compliant with the IEEE-488.2 standard. The command list and structure is identical to the RS-232 command set. IEEE-488.2 device address may be selected from 0 - 31. Factory default is 0.

IEEE Standard Codes, Formats, Protocols, and Common Commands (IEEE Std 488.2-1992) provides a detailed description of the IEEE common commands (identifiable in the command list by the asterisk as the first character.)

6. Theory of Operation

The TM-620 is calibrated for specific sensor excitation currents at the factory to optimize accuracy and minimize offsets and noise. Readings are taken via four-wire measurements, filtered, and sent at a 2Hz update rate to the front panel for display. Four wire measurements have the benefit of cancelling out the effects of lead resistance, which ensures that there are no offsets when performing RTD measurements. A 24bit ADC and stable, low noise amplifier stage ensure precise readings. Additional ADC calibration occurs after the device has been powered on for fifteen minutes to account for offsets caused by component and board temperature changes. Linear interpolation is used to calculate temperature data from stored curves.

7. Limited Warranty Policy

Cryomagnetics, Inc. warrants its products to be free from defects in materials and workmanship. This warranty shall be effective for two (2) years after the date of shipment from Cryomagnetics. Cryomagnetics reserves the right to elect to repair, replace, or give credit for the purchase price of any product subject to warranty adjustment. Return of all products for warranty adjustment shall be FOB Oak Ridge, TN, and must have prior authorization for such return from an authorized Cryomagnetics, Inc. representative.

This warranty shall not apply to any product which has been determined by Cryomagnetics, Inc. inspection to have become defective due to abuse, mishandling, accident, alteration, improper installation, or other causes. Cryomagnetics, Inc. products are designed for use by knowledgeable, competent technical personnel.

In any event, the liability of Cryomagnetics, Inc. is strictly limited to the purchase price of the equipment supplied by Cryomagnetics, Inc. Cryomagnetics, Inc. shall not assume liability for any consequential damages associated with use or misuse of its equipment.

Appendix A: Computer Interface Command Reference

Commands are separated into two categories: commands that are always available, which are generally queries, and commands which are only available when the device is in remote mode. Note that some commands are inhibited while the device is in the main menu. All commands that elicit a response from the instrument end with a question mark. All commands must be terminated with a return character ("\r"). The general command format is

<command block><RETURN>

where a command block is formatted as

<Command ><SPACE><Parameter>

Multiple commands can be sent at the same time by separating them with semicolons:

<command block 1>;<command block 2>;<command block 3><RETURN>

Responses to each subcommand are also separated by semicolons. For example, sending

**IDN?;UNITS V;UNITS?<RETURN>*

would return

Cryomagnetics,TM-622,2002,1.00;V<RETURN><LINEFEED>

Commands

The following table lists TM-620 commands, shows the mode where the command may be used, and provides a short command description. Command details are provided in the reference that follows.
Note: **For commands and queries that operate on a specific subchannel, the subchannel first needs to be selected using the "subch" command.**
Note: For commands and queries with an Availability of "Remote," the device must be in REMOTE mode for the command or query to operate. This can be done by sending a "Remote" command while the TM-620 display is at the top level (i.e. not in a menu).

Command	Availability	Description
CURVE?	Always	Returns the number of the currently selected curve
CURVE	Remote	Selects the curve to use for temperature conversions
ERROR?	Always	Set error response mode for RS-232 interface
ERROR	Remote	Query error response mode
EXC?	Always	Returns the excitation type
EXC	Remote	Set the excitation type
H-ALM?	Always	Returns the high alarm value in Kelvin
H-ALM	Remote	Sets the high alarm value in Kelvin
L-ALM?	Always	Returns the low alarm value in Kelvin
L-ALM	Remote	Sets the low alarm value in Kelvin
MEAS?	Always	Returns the reading
NAME?	Always	Returns the selected subchannel's name
NAME	Remote	Set the selected subchannel's name
SUBCH?	Always	Returns the currently selected subchannel
SUBCH	Remote	Sets the subchannel
UNITS?	Always	Returns selected units
UNITS	Remote	Sets units
LOCAL	Always	Puts the device into Local control mode
REMOTE	Always	Puts the device into Remote control mode
RWLOCK	Remote	Puts the device into Remote control mode with front panel lock
STAT?	Always	Returns the status of each channel
*CLS	Always	Clear Status Command
*ESE?	Always	Standard Event Status Enable Command
*ESE	Always	Standard Event Status Enable Query
*ESR?	Always	Standard Event Status Register Query
*IDN?	Always	Identification Query
*OPC?	Always	Operation Complete Command
*OPC	Always	Operation Complete Query
*RST	Always	Reset Command
*SRE?	Always	Service Request Enable Command
*SRE	Always	Service Request Enable Query
*STB?	Always	Read Status Byte Query
*TST?	Always	Self-Test Query
*WAI	Operate	Wait-to Continue Command

Table 3: Remote Interface Commands

Command Reference

This section describes how each command is used. The command syntax sections show required elements in <*angle brackets*> and optional parameters in *[square brackets]*. All numbers are decimal.

CURVE? Returns the current temperature curve

Availability: Always

Command Syntax: CURVE?

Response: <Curve Index>

Response Example: 3 **Response Range:** 1:8

Description: Returns the current subchannel's temperature curve index. Indices 1-6 are used for custom user curves, while indices 7 and 8 are used for PT100 and RO600 curves, respectively.

CURVE Select a temperature curve

Availability: Remote Mode

Command Syntax: CURVE <Channel Number>

Example: CURVE 2

Default Parameter: 1 **Parameter Range:** 1:8

Description: Sets the current subchannel's temperature curve index. Indices 1-6 are used for custom user curves, while indices 7 and 8 are used for PT100 and RO600 curves, respectively.

ERROR? Query error response mode

Availability: Always

Command Syntax: ERROR?

Response: <Error Mode>

Response Example: 0 **Response Range:** 0 or 1

Description: The **ERROR?** query returns the selected error reporting mode.

ERROR Set error response mode for RS-232 interface

Availability: Operate Mode

Command Syntax: ERROR <Error Mode>

Example: ERROR 1

Parameter Range: 0 or 1 (0 - disable error reporting, 1 - enable error reporting)

Description: The **ERROR** command enables or disables error messages when a serial interface is used. It is much easier to handle errors under program control when using a serial interface if error messages are disabled, but it is desirable to enable error messages if a terminal program is used to interactively control and query the TM-620.

EXC? Query the excitation mode

Availability: Always

Command Syntax: EXC?

Response: <Excitation>

Response Example: 1mA **Response Range:** 1mA, 100uA, 10uA, Disable

Description: The **EXC?** query returns the current subchannel's excitation mode.

EXC Set the excitation mode

Availability: Remote Mode

Command Syntax: EXC [Excitation Mode]

Example: EXC 100uA

Default Parameter: 1mA **Parameter Range:** 1mA, 100uA, 10uA, Disable

Description: The **EXC** sets the current subchannel's excitation mode.

H-ALM? Query high alarm threshold

Availability: Always

Command Syntax: H-ALM?

Response: <Degrees Kelvin>

Response Example: 105 **Response Range:** 0:1000

Description: The **H-ALM?** query returns the alarm high threshold temperature in Kelvin. If the measured temperature exceeds this threshold, the high alarm will trigger.

H-ALM Set high alarm threshold

Availability: Remote Mode

Command Syntax: H-ALM [Degrees Kelvin]

Example: H-ALM 105

Default Parameter: 200 **Parameter Range:** 0:1000

Description: The **H-ALM** command sets the alarm high threshold temperature in Kelvin. If the measured temperature exceeds this threshold, the high alarm will trigger.

L-ALM? Query low alarm threshold

Availability: Always

Command Syntax: L-ALM?

Response: <Degrees Kelvin>

Response Example: 15 **Response Range:** 0:1000

Description: The **L-ALM?** query returns the alarm low threshold temperature in Kelvin. If the measured temperature exceeds this threshold, the low alarm will trigger.

L-ALM	Set low alarm threshold		
Availability:	Remote Mode		
Command Syntax:	L-ALM [Degrees Kelvin]		
Example:	L-ALM 15		
Default Parameter:	10	**Parameter Range:**	0:1000

Description: The **L-ALM** command sets the alarm low threshold temperature in Kelvin. If the measured temperature exceeds this threshold, the low alarm will trigger.

MEAS?	Returns the current measurement.		
Availability:	Always		
Command Syntax:	MEAS?		
Response:	<Value> <Units>		
Response Example:	55K		

Description: Returns the most recent measurement and its units.

NAME?	Query subchannel name		
Availability:	Always		
Command Syntax:	NAME?		
Response:	<Name>		
Response Example:	1ST STAGE	**Response Range:**	A:Z, 0:9, <SPACE>

Description: The **NAME?** query returns the name of the selected subchannel.

NAME	Set subchannel name		
Availability:	Remote Mode		
Command Syntax:	NAME [Desired Name]		
Example:	NAME 2ND STAGE		
Default Parameter:	N/A	**Parameter Range:**	A:Z, 0:9, <SPACE>

Description: The **NAME** command sets the name of the selected subchannel, up to 9 characters in length. Names which are longer than 9 characters will be truncated.

SUBCH?	Query the current remote subchannel		
Availability:	Always		
Command Syntax:	SUBCH?		
Response:	<Subchannel>		
Response Example:	B	**Response Range:**	A, B, C, D

Description: Returns the currently selected subchannel. Any queries or commands which operate on a specific subchannel will operate on this selected channel.

SUBCH Set the remote subchannel

Availability: Remote Mode

Command Syntax: SUBCH <Subchannel>

Example: SUBCH A

Default Parameter: A **Parameter Range:** A, B, C, D

Description: Sets the currently selected subchannel. Any queries or commands which operate on a specific subchannel will operate on this selected channel. A TM-622 will have valid subchannels of A and B. A TM-624 will have additional C and D subchannels.

UNITS? Query selected units.

Availability: Always

Command Syntax: UNITS?

Response: <Selected Units>

Response Example: V **Response:** V, ohm, K, C, F

Description: Returns the current subchannel's units.

UNITS Select units.

Availability: Remote Mode

Command Syntax: UNITS <Unit Selection>

Example: UNITS C

Parameter Range: V, ohm, K, C, F

Description: Sets the current subchannel's units.

LOCAL Puts the device into Local control mode

Availability: Always

Command Syntax: LOCAL

Description: Prevents the device from receiving remote commands until the user manually changes the device back to non-Local mode via the front panel.

REMOTE Puts the device into Remote control mode

Availability: Always

Command Syntax: REMOTE

Description: Places the device in remote mode, which allows for commands to be sent to the device. Will only work when the device is in the top level display. The command will return an error when sent to the device while the device is in the main menu.

RWLOCK Puts the device into Remote control mode with front panel lock

Availability: Remote Mode (RS-232 Only)

Command Syntax: RWLOCK

Description: The **RWLOCK** command takes control of the TM-620 via the remote interface. All buttons on the front panel are disabled including the Local button. This command will be rejected if the menu system is being accessed via the front panel or if LOCAL has been selected via the Local button on the front panel. Pressing the Local button again when the menu is not selected will allow this command to be executed. The RWLOCK command is only necessary for RS-232 operation since the IEEE-488 RL1 option provides for bus level control of the Remote and Lock controls.

STAT? Query instrument detailed status

Availability: Always

Command Syntax: STAT?

Response: <Ch 1 Detailed Status>,<Ch 2 Detailed Status>,<Menu Mode>

Response Example: 8,3,0 **Response Range:** 0,0,0 to 127,127,1

Description: The **STAT?** query returns detailed instrument status as decimal values, and the status of local menu selection. When an operator selects the Menu, the instrument is taken out of operate mode, and <Menu Mode> is returned as 1. <Menu Mode> is returned as 0 when in operate mode. Channel detailed status is returned as a decimal number where each bit indicates a status condition of the channel. The meaning of each bit when set to one is shown:

Channel Detailed Status:

```
0   0   X   X   X   X   X   X
        |   |   |   |   |   |_____  Subchannel A/C overrange
        |   |   |   |   |_________  Subchannel B/D overrange
        |   |   |   |_____________  Subchannel A/C open
        |   |   |_________________  Subchannel B/D open
        |   |_____________________  Subchannel A/C alarm exceeded
        |_________________________  Subchannel B/D alarm exceeded
```

***CLS** Clear Status Command

Availability: Always

Command Syntax: *CLS

Description: The ***CLS** command operates per IEEE Std 488.2-1992 by clearing the Standard Event Status Register (ESR) and resetting the MAV bit in the Status Byte Register (STB).

***ESE?** Standard Event Status Enable Query

Availability: Always

Command Syntax: *ESE?

Response: <ESE Mask>

Response Example: 255 **Response Range:** 0 to 255

Description: The ***ESE?** command operates per IEEE Std 488.2-1992 by returning the mask set in the Standard Event Status Enable Register (ESE) by a prior *ESE command.

***ESE** Standard Event Status Enable Command

Availability: Always

Command Syntax: *ESE <mask>

Example: *ESE 255

Default Parameter: 0 **Parameter Range:** 0 to 255

Description: The ***ESE** command operates per IEEE Std 488.2-1992 by setting the specified mask into the Standard Event Status Enable Register (ESE).

***ESR?** Standard Event Status Register Query

Availability: Always

Command Syntax: *ESR?

Response: <Standard Event Status Register>

Response Example: 128 **Response Range:** 0 to 255

Description: The ***ESR?** command operates per IEEE Std 488.2-1992 by returning the contents of the Standard Event Status Register and then clearing the register.

<u>Status Byte Bit Allocations</u>:

```
X   X   X   X   X   X   0   1
|   |   |   |   |   |   |   |_____  Operation Complete
|   |   |   |   |   |   |_________  Request Control
|   |   |   |   |   |_____________  Query Error
|   |   |   |   |_________________  Device Dependent Error
|   |   |   |_____________________  Execution Error
|   |   |_________________________  Command Error
|   |_____________________________  User Request
|_________________________________  Power On
```

***IDN?** Identification Query.

Availability: Always

Command Syntax: *IDN?

Response: <Manufacturer>,<Model>,<Serial #>,<Firmware Level>

Response Example: Cryomagnetics,TM-624,8009,1.01

Serial # Range: 0001 to 65535 **Firmware Level Range:** 1.00 to 9.99

Description: The ***IDN?** command operates per IEEE Std 488.2-1992 by returning the TM-620 manufacturer, model, serial number, and firmware level.

***OPC?** Operation Complete Query

Availability: Always

Command Syntax: *OPC?

Description: The ***OPC** command operates per IEEE Std 488.2-1992 by placing an ASCII character "1" in the output queue since the TM-620 does not defer any commands for later processing.

***OPC** Operation Complete Command

Availability: Always

Command Syntax: *OPC

Description: The ***OPC** command operates per IEEE Std 488.2-1992 by placing the Operation Complete message in the Standard Event Status Register (ESR). The TM-620 processes each command as it is received and does not defer any commands for later processing.

***RST** Reset Command

Availability: Always

Command Syntax: *RST [hw]

Default Parameter: blank **Parameter Range:** blank or hw

Description: The ***RST** command operates per IEEE Std 488.2-1992 by returning the TM-620 to its power up configuration. This selects subchannel A as the default channel. If the optional parameter <hw> is provided, the instrument will perform a hardware reset one second later instead of returning to power up configuration as required by the Standard.

***SRE?** Service Request Enable Query

Availability: Always

Command Syntax: *SRE?

Response: <SRE Mask>

Response Example: 255 **Response Range:** 0 to 255

Description: The ***SRE?** command operates per IEEE Std 488.2-1992 by returning the mask set in the Service Request Enable Register (SRE) by a prior *SRE command.

***SRE** Service Request Enable Command

Availability: Always

Command Syntax: *SRE <mask>

Example: *SRE 255

Default Parameter: 0 **Parameter Range:** 0 to 255

Description: The ***SRE** command operates per IEEE Std 488.2-1992 by setting the specified mask into the Service Request Enable Register (SRE).

***STB?** Read Status Byte Query

Availability: Always

Command Syntax: *STB?

Response: <Status Byte>

Response Example: 65 **Response Range:** 0 to 255

Description: The ***STB?** command operates per IEEE Std 488.2-1992 by returning the Status Byte.

Status Byte Bit Allocations:

```
X   X   X   X   0   0   0   0
|   |   |   |
|   |   |   |___________________  MAV (Message Available)
|   |   |_______________________  ESB (Extended Status Byte)
|   |___________________________  MSS (Master Summary Status)
|_______________________________  Menu Selected
```

*** TST?** Self-Test Query

Availability: Always

Command Syntax: *TST?

Response: <Self test status>

Response Example: 1 **Response Range:** 1

Description: The ***TST?** command operates per IEEE Std 488.2-1992 by returning the self-test status. Explicit tests are not performed, but a 1 is returned for compliance with the specification.

*** WAI** Wait-to-Continue Command

Availability: Always

Command Syntax: *WAI

Description: The ***WAI?** command operates per IEEE Std 488.2-1992 by accepting the command without generating an error. Since the TM-620 only implements sequential commands the no-operation-pending flag is always TRUE.

Appendix B: Built In Temperature Curves

PT100

2.510000
4.260000
6.990000
10.490000
14.450000
18.100000
26.809999
35.349998
43.750000
52.040001
60.209999
68.300003
76.320000
84.269997
92.160004
100.000000
107.790001
115.540001
123.239998
130.889999
138.500000
146.059998
153.570007
161.039993
168.460007
175.830002
183.160004
190.429993
197.669998
204.850006
211.990005
219.080002
226.119995
233.119995
240.070007
246.979996
253.830002
260.649994
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